Weihao **Cheng**

3D PERCEPTION & MULTI-SENSOR FUSION · AUTONOMY · APPLIED ML · US CITIZEN

Lynnwood, WA 98087

(+1) 848-209-7004 | wcheng466@gmail.com | weihao-cheng

About me

Perception engineer and applied scientist working on 3D computer vision, multi-sensor fusion, and localization for autonomous systems. Trains and evaluates deep learning models on camera, thermal IR, LiDAR, magnetic, and inertial data, with a focus on cross-modal image matching, 3D reconstruction, and quantifying when a perception model can be trusted. Experience on ITAR-controlled government R&D programs.

Education

University of Illinois at Urbana-Champaign — *Urbana-Champaign, IL*

M.S. in Mechanical Science and Engineering — ***August 2021 - December 2024***

- Research group: Center for Autonomous Construction and Manufacturing at Scale (CACMS); Advisor: Dr. William (Bob) Norris
- Thesis: *Perception and mapping for unmanned ground vehicles in unstructured environments*
- Passed Doctoral Qualifying Exam (Specialization: Autonomy, Machine Learning, and Robotics)

Rutgers University - New Brunswick — *Piscataway, NJ*

B.S. in Mechanical Engineering (GPA: 3.92/4.0), Minor in **Economics** — ***August 2017 - May 2021***

- **Honors:** Summa Cum Laude (Class Rank 8/184), James J. Slade Scholar, Dean's List
- Contributed to securing **$38,722 NASA grant** for autonomous UAV research; authored technical reports and presented at **AIAA 2021** (3rd place)
- Conducted **UAV dynamics and computer vision** research with Dr. Bilgen on flight control and sensing platforms

Work Experience

Honeywell Aerospace — *Redmond, WA*

AI Engineer II (Project Lead) — *March 2025 - **Present***

- Reduced aerial-to-satellite image matching latency **15%** at equivalent accuracy by tuning **RoMa** coarse/fine resolutions
- Established error bounds for **FAA certification** of vision-based UAM landing by leading a **12-month Uncertainty Quantification** program
- Characterized perception error distributions across **500K+ aerial images** by building automated **Python/SQL** pipelines for statistical validation
- Modeled non-linear perception error to produce real-time confidence scores for **Kalman filter** sensor fusion
- Reduced accelerometer hysteresis loop area **10%** by training **RNN** models to compensate temperature-dependent biases
- Improved navigation performance in GPS-denied environments by reducing magnetometer noise **50%** via **U-Net**
- Enabled data-driven bid/no-bid decision by leading **Wisk Aero UAM RFI** sensor evaluation across **LiDAR, radar, EO, and IR** architectures
- Secured leadership approval to lead a new **LiDAR odometry** initiative through feasibility analysis and technical/cost assessment

Honeywell Aerospace — *Plymouth, MN*

Intern PhD AI Researcher — ***May 2024 - August 2024***

- Informed **UAM program bid strategy** by delivering **AI proof-of-concept** for vision-based landing directly to technical leadership
- Achieved **200% detection improvement** over classical baseline by prototyping **deep learning** marker detector with custom training pipeline
- Validated model against **UAM precision landing** standards via end-to-end **training, augmentation, and evaluation** pipeline

Mechanical Science and Engineering Department at UIUC — *Urbana-Champaign, IL*

Perception & Mapping Lead | **U.S. Army (CERL)** Autonomous Construction & Aerial Mapping — ***May 2022 - December 2024***

- Reconstructed large-scale **3D terrain** by registering **300+ aerial and ground LiDAR point cloud frames** via point cloud registration **(ICP)**
- Extended negative obstacle detection range **25%** by accumulating registered sparse point clouds over a sliding window of frames
- Built **multi-modal perception fusion pipeline** integrating LiDAR, stereo, and multispectral cameras for 2.5D/3D terrain mapping
- Trained **YOLOv8** on custom image datasets for negative obstacle detection in unstructured terrain
- Generated semantic traversability cost maps via **SLAM**, fusing elevation and terrain classification information from perception sensors
- Delivered **Perception & Mapping** workstream on time across 2-year Army program — managed roadmap, risk matrix, and stakeholder briefings

New Jersey Transit — *Newark, NJ*

Mechanical, Electrical, and Plumbing Engineer — ***June 2020 - August 2020***

- Reduced facility energy consumption by identifying anomalies in **HVAC sensor time-series** data
- Accelerated operational diagnostics by automating **time-series airflow** reporting workflows via Excel VBA scripting

Skills

3D Vision	Image matching, point cloud registration (ICP), SLAM, 3D reconstruction, SfM, NeRF, Gaussian Splatting, camera calibration
Sensor Fusion	LiDAR, stereo, thermal IR, radar, multispectral, IMU; Kalman filtering, uncertainty quantification
Deep Learning	PyTorch, U-Net, YOLOv8, LSTM, training/augmentation/evaluation pipelines, MLflow
Programming & Tools	Python (NumPy, SciPy, Pandas, Scikit-learn), SQL, ROS/ROS2, OpenCV, PCL, Docker, Git, Linux